Xuan Li Feng

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Education

MEng Biomedical Engineering, Imperial College London 2023 to 2027 (expected June 2027)

- Relevant Year 4 modules (in progress): Principles of Biomedical Imaging; Computer Vision and Pattern Recognition; Adaptive Signal Processing and Machine Intelligence; Reinforcement Learning for Bioengineers; Brain Machine Interfaces.

Research Experience

MEng Thesis: Stereo Near Infrared Fluorescence Imaging of Subsurface Vessels 2026 to 2027
Imperial College London | Supervisor: Dr Stuart Bowyer

- Built a pipeline that finds vessels in stereo near infrared fluorescence images, matches them between the two views and triangulates their depth.
- On 250 synthetic test pairs, a learned Siamese CNN matching cost gave a median depth error of 0.31 mm, compared with 0.56 mm for the best classical method (ZNCC).
- Simulating physically based light transport (Mitsuba 3) showed that vessels 2.5 to 5 mm deep appear only 1.1 to 1.5 mm deep, so scattering, not matching, limits depth. Phantom experiments are next.

Deep Neural Receptive Fields in Mouse Auditory Cortex 2025 to 2026
Imperial College London | Supervisor: Dr Andriy S. Kozlov | Project Leader and Manager

- Compared four spike prediction methods (STA, STC, maximum noise entropy and a five layer causal Conv1D network in PyTorch) on 46 neurons recorded during natural ultrasonic vocalisations.
- The Conv1D model had the highest median correlation (0.339 vs 0.290 with matched 50 ms bins), but was not significantly better than maximum noise entropy at its own 15 ms resolution ($p = 0.12$).
- Used Jacobian based dynamic receptive fields to measure gain change, temporal hold and shape change in each neuron.

Research Internship: Magnetoelastic Energy Harvesting June to August 2025
Zhejiang University | Principal Investigator: Professor Jikui Luo

- Worked full time on GaIn doped PVDF composites, using COMSOL, molecular dynamics and density functional theory to help interpret experiments. Coauthor of the resulting paper.

Selected Projects

Wearable Eye Tracking Communication Device for People with Cerebral Palsy 2024 to 2025
Imperial College London | Supervisor: Ian Radcliffe

- Designed the electrical schematic and PCB for a wearable device that combines eye tracking (OpenCV on a Raspberry Pi Zero 2 W), IMU head tilt commands and a Google Glass display. Also worked on the poster and final report.
- In testing, the device reached 78.2% speech output accuracy and a 91.8% IMU activation rate.

Remote Patient Monitoring System 2025
Imperial College London

- Wrote the Live Monitoring page and Digital Twin module of a Java Swing clinical monitoring system with a REST API and PostgreSQL database, including live waveform display.

Analogue Biomedical Instrumentation 2024 to 2026
Imperial College London

- Electronic stethoscope (pair project): two stage LM324 inverting amplifier with a total gain of 3700, confirmed in SPICE and on the breadboard; 20 to 500 Hz filtering of heart sounds and a Schmitt trigger driving an indicator diode on each heartbeat.
- Impedance respirometry system (individual lab report): stripboard circuit with 32 kHz current injection through skin electrodes, differential readout, Schottky diode envelope detection, RC band pass filtering and amplification; recovered the breathing waveform from a simulator set to 30 breaths per minute.

Publication

Lu, J., Zhang, K., Hazarika, D., Xu, L., Yao, J., **Xuan, L.**, … Luo, J. (2025). GaIn-Induced Polarization Control in PVDF for High-Efficiency Energy Harvesting and Instantaneous Wireless Sensing. *Energy & Environmental Materials*. doi:10.1002/eem2.70201

Technical Skills

Imaging and Vision	Stereo vision, 3D triangulation, vessel segmentation, centreline extraction, physically based rendering (Mitsuba 3), OpenCV, MR physics (coursework)
Machine Learning	PyTorch, Siamese CNNs, causal Conv1D networks, Jacobian based model interpretation, cross validation
Signal Processing and Statistics	Wiener and LMS filtering, AR modelling, spectral estimation, Cramér Rao bounds, hypothesis testing
Programming	Python, MATLAB, Java, JavaScript, Kotlin, Git, GitHub, Jupyter, PostgreSQL
Hardware	PCB design, analogue filtering and amplification, SPICE, Raspberry Pi, IMU sensing, COMSOL Multiphysics